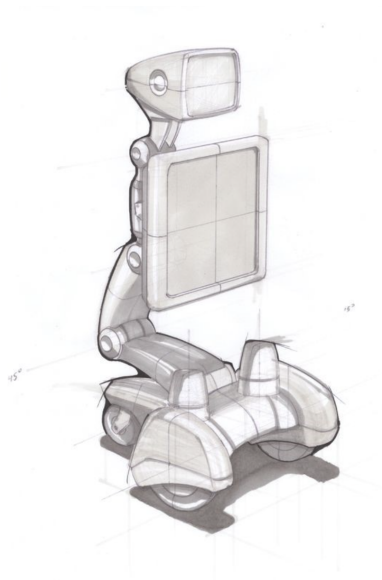


Broun Hall Telepresence Tour Guide

Capstone Design Proposal
Fall 2026
Electrical and Computer Engineering
Auburn University

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Concept Art by Beau Terrell

Executive Summary

The goal of this project is to build a mobile telepresence unit that can guide a local or remote visitor through the third floor of Broun Hall. The customer is the Auburn University Department of Electrical and Computer Engineering, represented by Dr. Thad Roppel. The intended end users are prospective students touring the department, students and visitors with mobility limitations who cannot easily walk the route, and remote guests joining a tour from off campus.

The proposed unit is a differential-drive mobile base carrying a camera and display on a height-adjustable mast. It can be driven either by a remote user through a browser client or by a local user through an onboard touchscreen, and it uses onboard obstacle sensors and a stored floor plan to avoid collisions and to show both users where it is.

Introduction

The Customer: The customer is the Auburn University Department of Electrical and Computer Engineering. Dr. Thad Roppel is the project sponsor and primary point of contact. The department will use the unit for recruiting tours, open houses such as E-Day, and remote visits.

The Problem Statement: Visitors who cannot be physically present in Broun Hall have no good way to tour the department. The department's previous telepresence robot showed that remote driving by itself is not enough. It had no onboard obstacle sensing, so remote users easily collided with people and objects; it had no stored map, so it could not tell its operator where it was and depended on a human guide walking alongside; and it had no local control mode, so even a person standing next to the unit had to log in remotely and drive through the resulting lag. Full autonomy has been ruled out by the sponsor as impractical for this application. The problem is therefore to design a human-driven mobile unit that is aware enough of its surroundings to be safe and easy to drive, that knows where it is on a stored map of the third floor, and that can be operated equally well by a remote visitor or by someone standing beside it.

Details

Alternative Approaches from Brainstorming:

- **Fixed 360-degree camera virtual tour:** Mount cameras at fixed points and let remote users switch views. Cheapest and lowest risk, but there is no mobility.
- **Retrofit a commercial mobile robot base:** Start from an off-the-shelf differential-drive base with published payload, encoders, and a supported software interface; add the mast, display, sensing, and telepresence software ourselves.
- **Guide Follower:** The unit tracks and trails a local guide rather than being driven. This removes the driving problem but still requires a guide, and gives a remote user no choice of where to go.
- **Custom chassis:** Design and build the drivetrain from scratch with departmental fabrication support. Most control over geometry and the most mechanical design content, at the cost of schedule.

Technical Challenges:

- **Indoor localization without GPS:** The unit must know its position on a stored floor plan to roughly 1 meter. The intended idea is to instrument the environment (RFID markers, wireless anchors, or a scanning rangefinder against a known map).
- **Network latency and coverage:** Video and control must cross the campus wireless network within a latency budget that keeps driving usable.
- **Mechanical constraints of the building:** Doorway widths, thresholds, floor-covering transitions, and hallway clearance set hard limits on the chassis.
- **Geometry:** A tall mast raises the center of gravity and creates a tipping risk that must be analyzed.
- **Detecting difficult obstacles:** Glass doors and partitions, stairwells and other drop-offs, low objects below the sensor band, and moving people are all cases a single sensor type will miss.

Resources Needed:

- Sponsor funding and materials support from the ECE department.
- Departmental machine shop and mechanical fabrication support.
- Access to the third floor of Broun Hall for route survey and testing after hours.

Potential SME's

- **Dr. Thad Roppel, ECE:** project sponsor; direct experience with the department's previous telepresence unit and its shortcomings.
- **Auburn GAVLAB:** navigation, state estimation, and sensor fusion.
- **Aarav Shah:** Created telepresence software over the summer with Dr. Roppel; his work may be implemented into the TourBot.
- **Professors with Labs on Floor 3 of Broun Hall:** Can indicate the proper information the TourBot should give visitors concerning their labs.

Work Breakdown Structure

